

A Technical Introduction to Lasso regression

$$L(\beta) = \|y - X\beta\|^2 + \lambda \|\beta\|_1$$

Section 1

Fused lasso In some cases, the phenomenon under study may have important spatial or temporal structure that must be considered during analysis, such as time series or image-based data. In 2005, Tibshirani and colleagues introduced the fused lasso to extend the use of lasso to this type of data. The fused lasso objective function is

Section 2

$$\min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{2} \|y - X\beta\|^2 + \lambda \|\beta\|_1 \right\}$$

Section 3

Lasso can set coefficients to zero, while the superficially similar ridge regression cannot. This is due to the difference in the shape of their constraint boundaries. Both lasso and ridge regression can be interpreted as minimizing the same objective function

Section 4

In 2005, Zou and Hastie introduced the elastic net. When $p > n$ (the number of covariates is greater than the sample size) lasso can select only n covariates (even when more are associated with the outcome) and it tends to select one covariate from any set of highly correlated covariates. Additionally, even when $n > p$, ridge regression tends to perform better given strongly correlated covariates. The elastic net extends lasso by adding an additional ℓ_2 $\{\text{displaystyle } \ell^2\}$ penalty term giving

Section 5

Generalizations Lasso variants have been created in order to remedy limitations of the original technique and to make the method more useful for particular problems. Almost all of these focus on respecting or exploiting dependencies among the covariates. Elastic net regularization adds an additional ridge regression-like penalty that improves performance when the number of predictors is larger than the sample size, allows the method to select strongly correlated variables together, and improves overall prediction accuracy. Group lasso allows groups of related covariates to be selected as a single unit, which can be useful in settings where it does not make sense to include some covariates without others. Further extensions of

group lasso perform variable selection within individual groups (sparse group lasso) and allow overlap between groups (overlap group lasso). Fused lasso can account for the spatial or temporal characteristics of a problem, resulting in estimates that better match system structure. Lasso-regularized models can be fit using techniques including subgradient methods, least-angle regression (LARS), and proximal gradient methods. Determining the optimal value for the regularization parameter is an important part of ensuring that the model performs well; it is typically chosen using cross-validation.

Section 6

and therefore it is standard to work with variables that have been made zero-mean. Additionally, the covariates are typically standardized $(\sum_{i=1}^N x_i^2 = 1)$ $\{\text{textstyle } \sum_{i=1}^N x_i^2 = 1\}$ so that the solution does not depend on the measurement scale. It can be helpful to rewrite

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where $H \propto \{\text{displaystyle } H_{\alpha}\}$ is again the hard thresholding operator and $I \{\text{displaystyle } \mathbb{I}\}$ is an indicator function (it is 1 if its argument is true and 0 otherwise). Therefore, the lasso estimates share features of both ridge and best subset selection regression since they both shrink the magnitude of all the coefficients, like ridge regression and set some of them to zero, as in the best subset selection case. Additionally, while ridge regression scales all of the coefficients by a constant factor, lasso instead translates the coefficients towards zero by a constant value and sets them to zero if they reach it.

Section 8

Convex relaxation interpretation Lasso can also be viewed as a convex relaxation of the best subset selection regression problem, which is to find the subset of $\leq k$ $\{\text{displaystyle } \leq k\}$ covariates that results in the smallest value of the objective function for some fixed $k \leq n$ $\{\text{displaystyle } k \leq n\}$, where n is the total number of covariates. The " ℓ_0 norm", $\|\cdot\|_0$ $\{\text{displaystyle } \|\cdot\|_0\}$, (the number of nonzero entries of a vector), is the limiting case of " ℓ_p norms", of the form $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$ $\{\text{displaystyle } \|x\|_p = \left(\sum_{i=1}^n |x_i|^p\right)^{1/p}\}$ (where the quotation marks signify that these are not really norms for $p < 1$ $\{\text{displaystyle } p < 1\}$ since $\|\cdot\|_p$ $\{\text{displaystyle } \|\cdot\|_p\}$ is not convex for $p < 1$ $\{\text{displaystyle } p < 1\}$, so the triangle inequality does not hold). Therefore, since $p = 1$ is the smallest value for which the " ℓ_p norm" is convex (and therefore actually a norm), lasso is, in some sense, the best convex approximation to the best subset selection problem, since the region defined by $\|x\|_1 \leq t$

$\|\mathbf{x}\|_1 \leq t$ is the convex hull of the region defined by $\|\mathbf{x}\|_p \leq t$ for $p < 1$.

Section 9

$b_{\text{OLS}} = \begin{cases} (1 - \lambda / R^2) b_{\text{OLS}} & \text{if } \lambda \leq R^2 \\ 0 & \text{if } \lambda > R^2 \end{cases}$

Section 10

$\min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{N} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_2^2 \right\}$

Section 11

but with respect to different constraints: $\|\beta\|_1 \leq t$ for lasso and $\|\beta\|_2 \leq t$ for ridge. The figure shows that the constraint region defined by the $\|\cdot\|_1$ norm is a square rotated so that its corners lie on the axes (in general a cross-polytope), while the region defined by the $\|\cdot\|_2$ norm is a circle (in general an n-sphere), which is rotationally invariant and, therefore, has no corners. As seen in the figure, a convex object that lies tangent to the boundary, such as the line shown, is likely to encounter a corner (or a higher-dimensional equivalent) of a hypercube, for which some components of β are identically zero, while in the case of an n-sphere, the points on the boundary for which some of the components of β are zero are not distinguished from the others and the convex object is no more likely to contact a point at which some components of β are zero than one for which none of them are.